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A multi-image steganography: ISS

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Abstract

Unlike single-image steganography, the scheme of payload distribution on different images plays a pivotal role in the security performance of multi-image steganography. In this paper, a novel multi-image steganography scheme: image stitching sender (ISS) is proposed, which achieves optimal payload distribution by optimizing the stitching scheme of multi-cover-images. In the ISS scheme, we employ peak signal-to-noise ratio as the similarity evaluation metric for the stitched cover image and stego image. Besides, genetic algorithm is used to find the local optimal solution for the similarity, corresponding to a locally optimal multi-image steganographic stitching scheme. The experiment demonstrates that ISS exhibits enhanced anti-detection capabilities in comparison to other multi-image steganography schemes. Furthermore, when combined with non-additive embedding methods, the ISS can achieve a more substantial improvement in security compared to additive embedding methods.

Keywords Multi-image steganography, Single-image steganography, Non-additive steganography, Payload distribution

Introduction

Encryption and steganography are two primary techniques used to secure communication data. Specifically, encryption is a more widely used technique compared to steganography. However, encrypted data has distinct characteristics that make it stand out from plaintext, which can potentially attract the attention of attackers (Cheddad et al. 2010).

Text, images, audio, video, and other media can serve as carriers for steganography, with images being the most frequently employed. In this paper, we primarily concentrate on image steganography. In recent years, some new techniques (Rustad et al. 2022) have been introduced into the literature, such as non-additive distortion techniques

(Filler and Fridrich 2010; Li et al. 2015), adaptive methods (Holub et al. 2014; Holub and Fridrich 2012; Pevný et al. 2010) and generative adversarial networks (GAN) (Volkhonskiy et al. 2020; Zhang et al. 2018a; Martín et al. 2023).

Early studies focused on designing coding methods with high security and large steganography capacity, such as Least Significant Bit (LSB) steganography and F5 steganography (Steganalysis and Westfeld 2001). Furthermore, researchers began to study adaptive steganography for choosing embedding positions based on the characteristics of the carrier, such as wet paper codes (Fridrich et al. 2004), ZZW codes (Zhang et al. 2008c), genetic algorithm (Kanan and Nazeri 2014). And then, Filler et al. designed the syndrome-trellis codes (STCs) (Filler et al. 2011), which can infinitely approximate the minimum additive distortion limit under some distortion function. After the introduction of STCs, the research on additive distortion in steganography primarily focused on the design of distortion functions within the STCs framework. Conversely, in the realm of non-additive steganography schemes, Filler and Fridrich proposed the Gibbs construction (Filler and Fridrich 2010) to address non-additive distortions in steganography. Since then, the development of steganography has

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divided into two directions: additive steganography and non-additive steganography. The additive steganography is mainly designed based on STCs coding including Highly Undetectable stego (HUGO) (Pevný et al. 2010), Wavelet Obtained Weights (WOW) (Holub and Fridrich 2012), Spatial Universal Wavelet Relative Distortion (S-UNIWARD) (Holub et al. 2014), High-pass, Low-pass and Low-pass (HILL) (Li et al. 2014) and so on. Nevertheless, non-additive steganography contemplates the joint influence of pixel modifications on carrier distortion rather than independent effects, primarily classified into three categories: theoretical model (Filler and Fridrich 2010; Kanan and Nazeri 2014), modification principle (Li et al. 2015; Denemark and Fridrich 2015; Tang et al. 2015; Wang et al. 2020; Qin et al. 2019; Duan et al. 2023) and adversarial detection (Fridrich and Kodovský 2013; Sedighi et al. 2015; Zhang et al. 2018a; Tang et al. 2018; Bernard et al. 2019, 2020, 2021; Martín et al. 2023). These steganography schemes will be described in detail in Sect. 2.

With the increasing amount of data exchanged during secret communication, the traditional single-image steganography is no longer sufficient for secure data hiding. In this context, multi-image steganography has gradually become a new research hotspot, as it offers the potential to increase the amount of data that can be securely hidden into the cover images. The crucial technique of multi-image steganography lies in distributing the payload into the cover images. Many popular payload distribution schemes have been applied in multi-image steganography schemes (Ker 2006, 2007b, 2008; Ker and Pevný 2012; Zhao et al. 2018; Cogranne et al. 2017; Liao et al. 2020). The Uniform Payload Distribution (UPD) scheme is a basic multi-image steganography scheme, which distributes the payload equally to each image. In recent years, the representative multi images schemes are Detectability Limited Sender (DeLS), Distortion Limited Sender (DiLS), Image Merging Sender (IMS) (Cogranne et al. 2017) and Embedding Strategy Based on Image Texture Complexity (ES-ITC), Embedding Strategy Based on Distortion Distribution (ES-DD) (Liao et al. 2020). Based on the experiments conducted in Liao et al. (2020), the ES-DD and IMS schemes are regarded as the most secure multi-image steganography schemes. In the ES-DD scheme, the pixel distortion of each image in the image set is calculated using the distortion function of HILL, and secret messages are embedded based on the distortion value of pixels. The ES-DD scheme considers the carrier distortion at the pixel level. However it can only compute additive distortion but not non-additive distortion. The IMS scheme, on the other hand, overcomes this limitation and is more convenient to implement compared to the ES-DD. Furthermore, experiments have

shown that the IMS scheme can achieve almost the same level of security as the ES-DD scheme. However, the IMS scheme lacks explicit guidelines for the image stitching process, including the optimal stitching sequence of cover images, batch size and image size, which may lead to the inability to fully utilize the payload allocation capability of the embedding scheme.

To address the limitations of the ES-DD scheme in computing non-additive distortions and the lack of explicit guidance for image stitching in the IMS, the Image Stitching Sender (ISS) scheme is proposed in this paper. Firstly, the ISS scheme utilizes the genetic algorithm (GA) as the optimization function to determine the optimal stitching sequence of cover images. Secondly, it conducts ablation experiments to validate the optimal image size and batch size. It is worth noting that we do not use the classification results of the steganalysis algorithm when choosing the fitness function for the GA due to the high computational cost involved. As an alternative, we experimented with Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) as metrics for evaluating the similarity between the stitched cover image and the stego image. By comparing the effectiveness of the two approaches, we determined that PSNR is more suitable as the fitness function for the GA.

The main contributions of this paper are as follows:

- (1) We propose an optimal stitching scheme for multi-cover-images, which achieves the optimal stitching sequence of the cover image set, image size and batch size, thereby achieving the goal of optimal payload distribution for multi-cover-images. In this scheme, PSNR is employed as the evaluation metric for the similarity between the stitched cover image and stego image. Besides, the GA is used to find the local optimal solution for the similarity, corresponding to a locally optimal multi-image steganographic stitching scheme.
- (2) We utilize the GA to jointly optimize the stitching scheme, making ISS more effective in enhancing the security of non-additive steganography compared to additive steganography. The experimental results demonstrate that ISS can demonstrate better resistance to detection than the current state-of-the-art multi-image steganography in most cases.

The rest of this paper is organized as follows. In Sect. 2, we introduce the research related to the single-image steganography and multi-image steganography. In Sect. 3, we introduce the ISS multi-image steganography scheme, including the IMS module and the GA module. And we provide a detailed explanation of how the ISS works, Including encoding methods and fitness

calculation and how it performs selection, crossover, and mutation operations. In Sect. 4, we conduct five experiments aimed at determining the optimal fitness function, the optimal batch size and image size, and comparing the security and carrier's change rate of different multi-image steganography.

Related works

In this section, we present the currently popular single-image steganography in Subsect. 2.1, including additive steganography and non-additive steganography. In Subsect. 2.2, we introduce the development history and latest advances of multi-image steganography.

Single-image steganography

The purpose of steganography is to minimize the distortion caused by embedding operations on the carrier, so that the sender can hide secret messages in images for transmission without arousing suspicion from attackers (Fridrich and Filler 2007). Steganography techniques have been developed based on the fundamental theory that embedding the same secret messages into images with different texture features results in different degrees of distortion. Moreover, the basic problem that steganography techniques aim to solve is to use efficient and accurate methods to find the most suitable areas or pixels in the image for embedding secret messages, which enables us to achieve the minimal possible embedding distortion. Thus, many scholars have proposed content-adaptive steganography. These steganography can be classified into two categories based on the way of calculating distortion: additive distortion (Holub et al. 2014; Holub and Fridrich 2012; Pevný et al. 2010; Li et al. 2014) and non-additive distortion (Filler and Fridrich 2010; Li et al. 2015).

If we consider the distortion caused by the embedding operation of steganography as additive, which means that the modification to any pixel in the cover image does not affect the distortion value of other pixels. In this case, the distortion of each pixel is calculated separately, and the total embedding distortion is acquired by summing up the distortion values of each pixel. With the help of STCs, we can approach the theoretical minimum of the total distortion for a specific distortion function.

In recent years, more attention has been paid by scholars to the design of distortion functions due to the advent of STCs. The HUGO (Pevný et al. 2010) computes distortion by taking the weighted sum of differences between the feature vectors of the cover image and their corresponding vectors of the stego image, which is obtained by altering only the selected pixel. The WOW (Holub and Fridrich 2012) quantifies the predictability of pixels in the cover image and utilizes directional filters to embed more

secret messages in regions of the image that are harder to predict. Based on this, the S-UNIWARD (Holub et al. 2014) makes minor adjustments to the directional filters and computes the cost as the sum of relative changes in wavelet coefficients across all subbands. Experiments have shown that S-UNIWARD is slightly better than WOW in terms of security, and they both perform significantly better than HUGO. Furthermore, the HILL proposes further enhancements to WOW by utilizing a high-pass filter and two low-pass filters. This allows the steganography to more accurately identify less predictable areas using the high-pass filter and extend the low-cost property of complex textured areas to surrounding pixels using the low-pass filters.

Conversely, if we consider the distortion caused by the embedding operation of steganography as non-additive, then modifying a single pixel can affect the calculation of the distortion cost of the surrounding pixels. When computing non-additive distortion, we cannot simply calculate the sum of the modification costs of individual pixels. Instead, the non-additive distortion function needs to be updated iteratively or dynamically to determine the distortion value. Non-additive steganography can be classified into three categories based on the implementation process.

Firstly, the non-additive distortion function designed based on theoretical model. The most representative scheme is the Gibbs construction (Filler and Fridrich 2010), and we can simulate the interaction between embedding changes by representing macroscopic features as a sum of local feature functions. HUGO-Bounding Distortion (HUGO-BD) (Filler and Fridrich 2010) is a typical application of the Gibbs construction.

Secondly, in order to reduce the computational cost, scholars have begun to investigate heuristic non-additive distortion functions based on the modification principle. For example, Li et al. (2015) proposed Clustering Modification Directions (CMD) non-additive steganography for spatial images, while Denemark and Fridrich (2015) proposed Synchronization (SYNCH) non-additive steganography. Both CMD and SYNCH first divide the cover image into non-overlapping sub-blocks, and then dynamically calculate the distortion of the non-embedded parts according to the modification direction of the embedded part. However, the two methods differ in their approach to dividing the image and updating the distortion. CMD performs slightly better than SYNCH in terms of undetectability. Inspired by the embedding direction consistency principle, Tang et al. (2015) extended the CMD to color spatial images and proposed the CMD-C. Wang et al. (2020) proposed the GINA by considering the correlation between different color channels of spatial image. Qin et al. (2019) proposed the CPV by converting

a multi-channel pixel into a superpixel, so that the distortion can be adaptively assigned to the three color channels based on complexity and channel correlation.

Thirdly, we can design non-additive distortion functions based on adversarial detection. For example, the Multivariate Gaussian (MG) (Fridrich and Kodovský 2013) initializes cover modeling using quantized multivariate Gaussian models with different but independent distributions. The MG minimizes the KL divergence between cover and stego images. Minimizing the Power of Optimal Detector (Mi-POD) (Sedighi et al. 2015) is a typical adversarial steganography that designs the embedding cost by minimizing the detection ability of the optimal detector. Zhang et al. (2018a) proposed an adversarial sample steganography that can deceive deep learning-based steganalysis models by causing them to misclassify a stego image as a cover image. Tang et al. (2018) proposed the ADV-EMB, which is divided into two parts: one for conventional embedding and the other for adversarial embedding. Bernard et al. (2020) proposed an adversarial embedding optimization method based on a min-max strategy. Besides, they also developed a new strategy called Backpack (Bernard et al. 2021) to adjust distortion and approximated non-additive adversarial distortion as additive distortion to solve the distortion updating problem using gradient information. Tan et al. (2021) proposed CHAT-GAN, a novel end-to-end image steganography framework that integrates channel attention mechanisms to enhance stego image quality and message extraction accuracy. Additionally, it adopts an advanced steganalysis network with an improved high-pass filter in the preprocessing layer, offering enhanced security with reduced algorithmic complexity.

In single-image steganography, whether adopting additive or non-additive distortion, the differences of carriers is not considered, thus failing to achieve payload distribution among carriers. As a result, single-image steganography falls short of multi-image steganography in terms of security.

Multi-image steganography

Unlike single-image steganography, multi-image steganography differs in that it can reduce the average payload of the carriers by increasing the number of carriers. Additionally, it can adaptively distribute the payload into the carriers based on their characteristics. The concept of multi-image steganography was first proposed by Ker (2006) in 2006, who investigated methods for pooling steganalysis evidence to achieve more secure steganography. And then, in Ker (2007a), they proposed a threshold game for the steganalyzer and steganographer in order to evaluate the security of multi-image steganography. Additionally, they discovered that the security of

multi-image steganography is proportional to the square of the number of cover images in Ker (2007b). According to Ker (2008), it is safer to evenly distribute the payload among each cover image in multi-image steganography. In 2012, Ker and Pevny (2012) proposed the max-greedy strategy for distributing payload in multi-image steganography. In 2017, Zhao et al. (2018) developed a scheme that spreads the payload to images with larger sizes and more balanced histograms. Cogranne et al. (2017) compared the security of three payload distribution schemes: DeLS, DiLS, and IMS. Among them, DeLS and DiLS allow the steganographer to adjust the KL divergence and distortion between the cover image and the stego image, respectively; IMS merges all cover images into one large image and utilizes single-image steganography to distribute the payload.

In 2020, Liao et al. (2020) proposed two adaptive payload distribution schemes for multi-image steganography based on image texture features, ES-ITC and ES-DD. ES-ITC calculates the steganographic capacity of each cover image using its entropy and embeds secret messages in cover images with higher steganographic capacity as much as possible. ES-DD determines the embedding distortion of each pixel in the cover images using HILL's distortion function, and embeds secret messages in pixels with low distortion values as much as possible. According to Liao et al. (2020), ES-DD and IMS are comparable in guilty actors experiments, indicating similar security performance. Additionally, in the "out-of-bag" error rate $E(oob)$ experiment, the security of these three payload distribution schemes ES-ITC, ES-DD, and IMS is very similar.

The proposed multi-image steganography scheme: ISS

In this section, we first introduce the basic multi-image steganography scheme UPD in Subsect. 3.1. This scheme is equivalent to directly using single-image steganography to distribute the payload evenly in multi-image steganography. In Subsect. 3.2, we explain how the IMS module of our proposed ISS algorithm works. Then, in Subsect. 3.3, we will discuss the encoding method, fitness function and the three operations of GA: selection, crossover and mutation. We will also introduce the ISS framework, which is specifically designed to find the optimal stitching scheme for a set of cover images.

The basic scheme: uniform payload distribution

With the UPD scheme, the payload of each cover image remains the same. Thus, for a cover image set with n images, we can consider it as a single-image steganography operation that is repeated n times. Therefore, the

security of the UPD is equivalent to the security of the single-image steganography operation.

The size of the secret messages embedded in each image is equal with the UPD. The advantages of the UPD is that neither the sender nor the receiver need to have any additional secret key. The secret messages will be divided into n equal parts and embedded into n images from the cover image set. In our experiment, the embedding capacity of each image is determined by its pixel size, indicating that the maximum payload size is 1(bpp).

The IMS module of ISS

In the IMS module, we first merge all the images in the cover image set into a large rectangular image with dimensions $r \times s$, where r and s represent the number of rows and columns of the merged large rectangular image, respectively. To make the cover images as adjacent to each other as possible, we try to make the values of r and s as close as possible. If there are not enough images in the last row, we will randomly opt for several images from the existing cover images to fill the remaining space in the rectangle image.

After obtaining the large rectangular image, we will use it as the cover image for the single-image steganography scheme. The payload can be distributed into these images by relying on the steganography scheme. Once the secret messages has been embedded, the large stego image will be partitioned into $r \times s$ small images. These images can be transmitted to the receiver through a public channel.

The framework of IMS module is shown as Fig. 1, the sender randomly selects n images, which can be merged into a large image of size $r \times s$ using the IMS module. The secret data M with length $|M|$ is split into $r \times s$ pieces. The length of secret messages embedded into a single image in row i and column j can be formulated as $|m_{ij}|$. The relationship between $|M|$ and $|m_{ij}|$ is shown in Eq. 1.

$$|M| = \sum_{i=1}^r \sum_{j=1}^s |m_{ij}| \quad (1)$$

The value of $|m_{ij}|$ is determined by the payload distribution scheme, which can be formulated as $\Psi(\cdot)$. Let X represent the cover image set, and the features of images are formulated as IF_X , we can compute the m_{ij} by the Eq. 2.

$$[m_{11}, m_{12}, \dots, m_{rs}] = \Psi(M, IF_X) \quad (2)$$

Note that the stego image set Y can be acquired by using the cover image set X and the secret message m_{ij} of each image.

After acquiring m_{ij} , we can use the embedding schemes $Emb(\cdot)$ to generate the corresponding stego image y_{ij} and the stego image set Y as Eq. 3.

$$\begin{cases} Y = \{y_{11}, y_{12}, \dots, y_{rs}\} \\ y_{ij} = Emb(x_{ij}, m_{ij}) \end{cases} \quad (3)$$

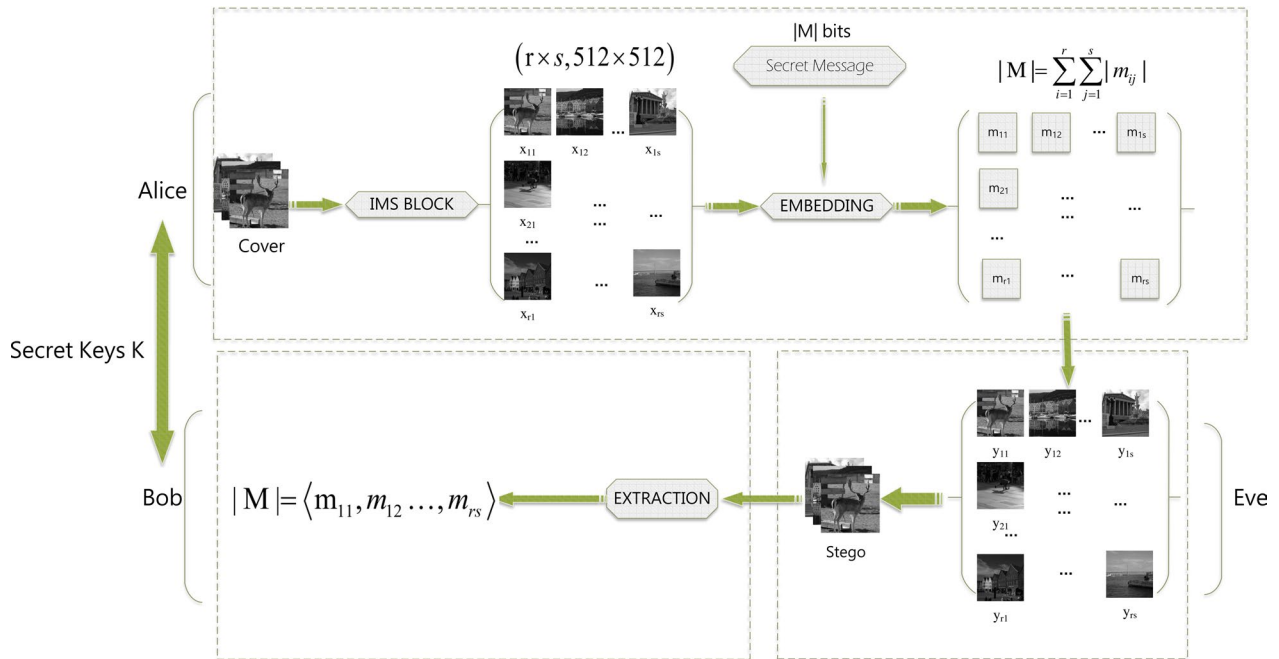


Fig. 1 The framwork of IMS

The receiver can get the stego images from the public channel and extract the secret messages from it. It is important to note that the extracted messages may be split and unordered. In order to splice an accurate and complete secret message, the receiver needs to know the length of the secret messages embedded in each image and the position $index(i, j)$ of each image which are contained in the secret key K . In our experiment, we assume that this key has already been shared by the communicating parties over a secure channel.

As pointed out in Ker and Pevný (2014), the allocation of payload might be part of a shared secret key, or stored in the first few bits of payload in a fixed position. In our work, we chose the first method, which ensures that the secret key K and the secret data M are independent of each other, resulting in higher security. According to the analysis in Liao et al. (2020), the conditional entropy of the secret key $H(K|Y, \Psi)$ must be greater than the conditional entropy of the secret data $H(M)$ to prevent an attack via K . The necessary uncertainty about secret data M must be at most the same size to make an attack on secret key K by the knowledge of stego image Y and payload distribution scheme Ψ , which can be formulated as Eq. 4..

$$H(K|Y, \Psi) = H(M) + H(K|M) > H(M) \quad (4)$$

The receiver can use the function $Ext(\cdot)$ to extract the message as Eq. 5:

$$\begin{cases} K = \{(p_1, index_1), (p_2, index_2), \dots, (p_{r \times s}, index_{r \times s})\} \\ Ext(\cdot) = Y \times K \rightarrow M \end{cases} \quad (5)$$

where p_i refers to the sub-payload of the i -th image among a group of $r \times s$ images, and $index_i$ represents the positional index of the i -th image in the concatenated large image. The exchange of the private key K between the sender and the receiver occurs after the secret messages has been embedded but before the stego image is sent. For each stego image y_{ij} , a corresponding key k_i is used to extract the secret message m_{ij} . The original secret data M can be acquired by concatenating all m_{ij} in the order of their index numbers. The extraction process can be expressed in Eq. 6.

$$\begin{cases} m_{ij} = Ext(y_{ij}, k_i) \\ M = \{m_{11}, m_{12}, \dots, m_{rs}\} \end{cases} \quad (6)$$

As pointed out in Liao et al. (2020), the cover image set X , the secret data M , payload distribution scheme $\Psi(\cdot)$, the embedding scheme $Emb(\cdot)$, the extracting scheme $Ext(\cdot)$

and the secret key K together constitute the multi-image steganography Γ , which can be formulated as Eq. 7.

$$\Gamma = \{X, M, \Psi(\cdot), Emb(\cdot), Ext(\cdot), K\} \quad (7)$$

The GA module of ISS

In our experiments, we find that different stitching sequences of the cover image set result in varying levels of detection resistance for multi-image steganography schemes. To identify the most secure stitching sequence, we utilized the genetic algorithm as the optimization method.

The solution space for the problem of finding the optimal stitching sequence of cover images is very large, containing $(r \times s)!$ solutions. In our experiments, we assume that the process of approximating the optimal solution is continuous and progressive, which enable us to use an iterative approach to find the optimal stitching sequence. The GA is capable of finding the optimal solution in a large solution space by utilizing operators such as selection, crossover, and mutation, which are inspired by the process of natural selection. By continuously evolving the initial population, we can get a near-optimal solution that is very close to the theoretical optimal solution.

we chose permutation encoding to generate the initial population in order to find the optimal image stitching sequence. Assuming that we randomly generate n individuals as the initial population, the primary population can be formulated as Eq. 8.

$$POP(primary) = \{Order_1, Order_2, \dots, Order_n\} \quad (8)$$

where $Order_i$ means the i -th sequence of cover image set, and each sequence corresponds to a pair of large images: $\hat{X}_i$ and $\hat{Y}_i$, $\hat{X}_i$ represents the large image that is concatenated together by the IMS module, while $\hat{Y}_i$ represents the large image that has already embedded the secret information. The relationship of $\hat{X}_i$ and $\hat{Y}_i$ can be formulated as Eq. 9.

$$\Gamma_{IMS}(\hat{X}_i) = \hat{Y}_i \quad (9)$$

Clearly, the more similar $\hat{X}_i$ and $\hat{Y}_i$ are, the more secure the steganography scheme will be. To identify the best individual from n sequences of X , it is necessary to employ an objective evaluation function that quantifies the similarity between $\hat{X}_i$ and $\hat{Y}_i$. In our experiment, we choose PSNR as the evaluation function. And we use $PSNR_i$ to represent the value of PSNR between the i -th cover image and the stego image. A higher PSNR value indicates a higher similarity between $\hat{X}_i$ and $\hat{Y}_i$, which implies a more secure steganography.

We consider PSNR as the fitness function of GA. Therefore, the problem of finding the optimal cover image order is transformed into the problem of finding the order that can achieve the highest PSNR value.

After generating the primary population, we are able to complete the initialization of the next generation population in two steps. Firstly, we choose the optimal individual $Order_{max}$ from the initial population and clone it t times. Secondly, we generate $n - t$ new individuals by applying the operations of selection, crossover, and mutation to the primary population. The second step consists of three sub-steps: selection, crossover, and mutation.

(1) Selection

We have implemented the Stochastic Universal Sampling (SUS) method as our selection operator. With the SUS operator, individuals in the population are mapped onto a probability interval of 100%, as shown in Fig. 2. The length of the segment corresponding to each individual is proportional to its fitness. By using SUS, we can maintain diversity in the offspring population while also retaining elite individuals. A set of equidistant pointers are randomly pointed to the above segment, and the individual indicated by each pointer is the individual to be selected. The distance between Pointers $Dist$ and the distance of the first pointer from the origin $Dist'$ can be formulated as Eq. 10.

$$\begin{cases} Dist = \frac{1}{N_{sel}} \\ Dist' = random(0, Dist) \end{cases} \quad (10)$$

where the N_{sel} means the number of individuals to be selected, and we set the N_{sel} as $n - t$. the $Dist'$ could be any value from 0 to $Dist$. As we can see from the Fig. 2, the larger N_{sel} is, the smaller $Dist$ is, which means the diversity of individuals is stronger. So, determining an appropriate N_{sel} is one of the key parameters of the GA.

The operation of selection $Sel(\cdot)$ can be formulated as Eq. 11.

$$\begin{aligned} Sel(\cdot) &= sus(POP(primary), \Gamma_{IMS}(\cdot), SIM(\cdot), N_{sel}) \\ &= Selchrom\{order_1', order_2', order_{n-t}'\} \end{aligned} \quad (11)$$

where the $sus(\cdot)$ means the selection operator of SUS, the $SIM(\cdot)$ represents the operation of using PSNR as a metric to assess the similarity between cover images and stego images, and the $Selchrom$ means the chromosome of $n - t$ individuals, which can be considered as the input of the operation of crossover.

(2) Crossover

The crossover operation is a highly effective measure for increasing population diversity. This operation enables the generation of new individuals by combining the genetic information contained in the paternal chromosomes of both parents. It should be noted that the choice of an appropriate crossover operation depends on the encoding method of the chromosome. In our experiment, we are dealing with a combinatorial optimization problem where each bit of the chromosome must be unique, corresponding to the index value of the cover image during the merging process. Thus, we adopt permutation to encode as the method for encoding chromosomes within the GA.

To perform chromosome crossover in combinatorial optimization problems, we have opted for the partially matched crossover algorithm (PMX) (Khan 2015) as our method of crossover. This method is similar to the value exchange recombination method shown in Fig. 3, but with one key difference: with PMX, we have to ensure that the offspring chromosomes generated after the crossover maintain the characteristics of permutation coding, meaning that each bit in the chromosome must be unique. To achieve this, we have added a step after the crossover operation where we exchange duplicated genes

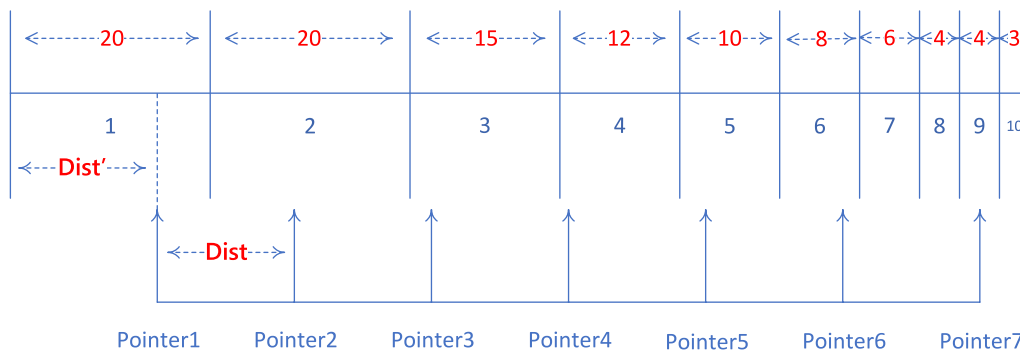


Fig. 2 An example of the SUS method

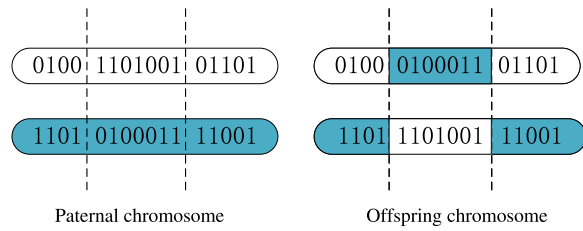


Fig. 3 The method of values exchanged recombination

in the offspring chromosome. The procedure of the PMX method is shown in Fig. 4.

In addition, we need to set the probabilities of chromosome crossover $Pr(Chrom_c)$ and segment crossover $Pr(Seg_c)$. Among them, $Pr(Chrom_c)$ determines how many chromosomes will be selected from the primary population for crossover, while $Pr(Seg_c)$ represents the

probability that each gene in the selected chromosome will be crossed over. This can also be described as the length of the segment being crossed.

In practice, since the fitness of each chromosome in the previous generation has been calculated and the fittest individual has been retained in the initial step, we choose to perform crossover on every chromosome, with $Pr(Chrom_c)$ set at 100%. Furthermore, to test the impact of modifying different segments on fitness as much as possible, we set $Pr(Seg_c)$ to 70%, which means that 70% of each chromosome will be crossed over. The crossover operation $Cros(\cdot)$ can be expressed as Eq. 12.

$$\begin{aligned}
 Cros(\cdot) &= PMX(Selchrom, Pr(Chrom_c), Pr(Seg_c)) \\
 &= Croschrom\{order_1', order_2', order_{n-t}'\} \quad (12)
 \end{aligned}$$

where the $Croschrom$ means the $n - t$ chromosomes which have been crossed over.

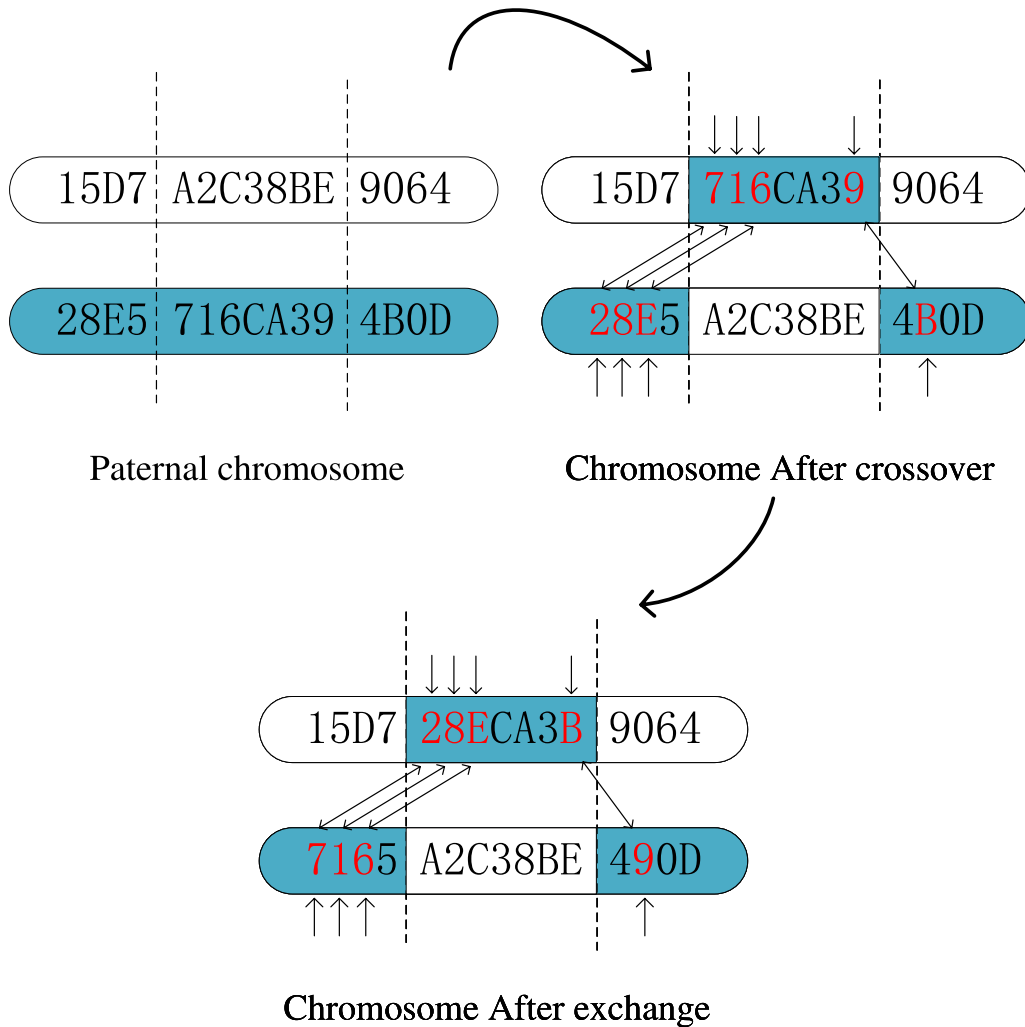


Fig. 4 The method of PMX

(3) Mutation

Mutation is the process of generating a new chromosome by altering a portion of its elements. This process enhances the diversity of the population and reduces the risk of the evolutionary algorithm being trapped in local optima. In accordance with the definition of crossover probability, we designate the probability of chromosome mutation as $Pr(Chrom_m)$, set at 100%, and the probability of segment mutation at 70%. Considering the characteristics of permutation encoding, we have chosen the Two Point Swapping (TPS) method as our mutation operator. The procedure of the TPS method is shown in Fig. 5. Unlike crossover operations that typically involve two parent chromosomes, mutation operations frequently involve only one parent chromosome. Specifically, we select consecutive or discontinuous segments of the same length in the parent chromosome, and then swap their positions to obtain a new offspring chromosome.

The operation of mutation $Mut(\cdot)$ can be formulated as Eq. 13.

$$\begin{aligned} Mut(\cdot) &= TPS(Croschrom, Pr(Chrom_m), Pr(Seg_m)) \\ &= Mutchrom\{order_1', order_2', order_{n-t}'\} \end{aligned} \quad (13)$$

where the $Mutchrom$ means the $n - t$ chromosomes which have been mutated.

After completing these two steps, we are able to get t chromosomes representing the best fitness and $n - t$ chromosomes resulting from selection, crossover, and mutation. These two parts are then concatenated to form the second generation population of the GA. The formulation of this process is as Eq. 14.

$$POP(2^{nd}) = Concatenate(t \times Order_{max-1}, Mutchrom) \quad (14)$$

We can generate G generations of populations using the same way that we formed the second generation population. From each generation, we can select an individual $Order_{max-i}$ with optimal fitness $PSNR_{max-i}$. We can then acquire the large stego image $\widehat{Y}_{max-i}$ of the i -th generation based on $Order_{max-i}$ as Eq. 15.

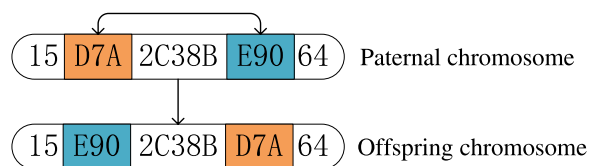


Fig. 5 The method of TPS

$$\begin{cases} \widehat{X}_{max-i} = merge(X, Order_{max-i}) \\ \widehat{Y}_{max-i} = \Gamma_{IMS}(\widehat{X}_{max-i}) \end{cases} \quad (15)$$

where the $\widehat{X}_{max-i}$ means the large image merged from X by the $Order_{max-i}$, and the $\widehat{Y}_{max-i}$ means the corresponding stego large image. The value of i can be $\{1, 2, \dots, G\}$.

Then we can obtain $PSNR_{max}$ as well as its corresponding chromosome $Order_{max}$ by comparing the similarity between $\widehat{X}_{max-i}$ and $\widehat{Y}_{max-i}$ for each generation as Eq. 16.

$$\begin{aligned} Order_{max} &= argmax(PSNR_{max-i} : f(Order_{max-i})) \\ &= argmax(SIM(\widehat{X}_{max-i}, \widehat{Y}_{max-i})) \end{aligned} \quad (16)$$

The payload distribution scheme $\Gamma(ISS)$ and the steganography $\Psi(ISS)$ can be formulated as Eq. 17.

$$\begin{cases} \Psi(ISS) = \{\Gamma(IMS), Order_{max}, X\} \\ \Gamma(ISS) = \{X, M, \Psi(ISS), Emb(\cdot), Ext(\cdot), K\} \end{cases} \quad (17)$$

As shown in Fig. 6, the ISS multi-image steganography scheme can be summarized in the following five steps:

- (1) Randomly generate n permutation coding chromosomes. Each of these chromosomes indicates the sequence in which the images are merged. These n chromosomes are considered as the primary population for the GA;
- (2) Reorder the input images according to the chromosomes and generate n groups of images. Merge each group into a large image using the IMS module. Mark each large image as a cover image and embed the secret messages into it. We can then get n pairs of cover image and stego image;
- (3) Calculate the similarity of each pair of cover and stego images using the PSNR metric. The chromosome with the highest PSNR value is chosen as the optimal individual;
- (4) Clone the optimal individual t times and select $n - t$ individuals from the population based on the PSNR of each pair of cover image and stego image. Combine the t optimal clones with the $n - t$ selected individuals to form the next generation population;
- (5) Repeat (2),(3),(4) for G times and record the optimal individual of each generation. The chromosome with the highest PSNR is selected as the output of the ISS scheme.

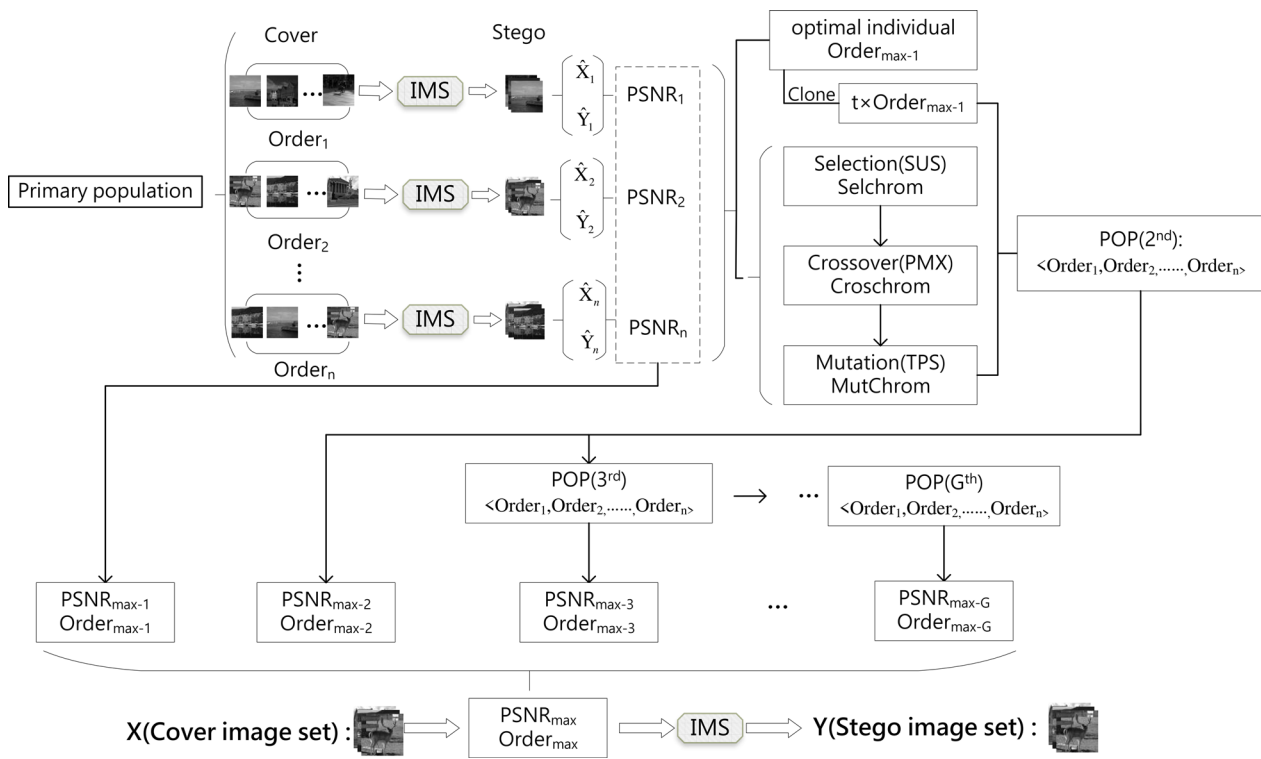


Fig. 6 The framework of ISS

Experimental results

In this section, we conduct several experiments to determine the best parameters for the ISS multi-image steganography scheme. We also aim to demonstrate its superiority in the multi-image steganography task. In Subsect. 4.1, we determine the optimal genetic iterations, population size for each generation, and the most effective fitness function for the GA. In Subsect. 4.2, we obtain the optimal image size and batch size for multi-image steganography scheme through ablation experiment. In Subsect. 4.3, we compare the security performance of ISS with the UPD, IMS and ES-DD schemes. In Subsect. 4.4, we investigate the impact of different image stitching schemes on the entropy of the merged image's gray-level co-occurrence matrix (GLCM) (Tamura et al. 1978). In Subsect. 4.5, we analyze the characteristics of different multi image steganography scheme by examining the change rates of each image in the cover image set.

In this paper, we conduct UPD, ES-DD, IMS, and ISS along with three single-image steganography (S-UNIWARD, WOW, HUGO-BD) to embed secret information into a cover image set. We select the cover images from the BOSSbase 1.01 dataset, which contains 10,000 grayscale images of size 512×512 .

Determine the optimal parameters for the GA

In this subsection, we compare the performance of the GA with different numbers of iterations and population sizes to determine the optimal parameters. In addition, when choosing the fitness function for the GA, we consider two metrics that can evaluate image similarity: PSNR and SSIM.

In our experiment, we assume the presence of an omniscient detector, which means that the detector is aware of the multi-image steganography scheme, embedding scheme, and the total payload of the steganography. To assess the security of the steganography, we employed the Spatial Rich Model (SRM) method (Fridrich and Kodovsky 2012) to extract 34,671-dimensional features from both the cover images and the stego images.

We randomly select 10 sets of image data from the BOSSbase 1.01 dataset, with each set containing a total of 512 images. Half of the images are used for training, and the other half are used for testing. The ISS is employed to embed the same secret messages into each pair of training and testing sets. The population scale, denoted as n , was set as {50, 100}, and the number of generations G was set as 50. We record the chromosomes of individuals with the best fitness in each generation. In each experiment

group, we also record the best individuals obtained by the GA at iterations 10, 20, 30, 40, 50.

We use the training cover image set, testing cover image set, and their respective stego image set as input data for the SRM steganalysis method. We assess the security of the multi-image steganography under different parameters by calculating the sum of the false positive rate and the false negative rate, which is referred to as the ensemble's "out-of-bag" error rate. This error rate is denoted as $E(oob)$. A higher $E(oob)$ indicates better steganographic security.

Tables 1 and 2 shows the $E(oob)$ metric of the ISS scheme when using PSNR and SSIM as the fitness function, respectively. In the table, R represents the ranking of the best individual's fitness at the corresponding generation among the GA's 50 generations. Conversely, n

denotes the number of individuals in each generation of the population, and G represents the number of generations. Each value of $E(oob)$ and R is given by the average of 10 experimental groups. All the steganography experiments are embedded with payload of 0.2bpp.

Based on the data in Tables 1 and 2, we can observe that the ISS scheme performs slightly better when the population size parameter n is set to 50 than when it is set to 100. Additionally, increasing the number of generations in the GA does not necessarily lead to better results. As shown in Fig. 7, the highest value of $E(oob)$ is achieved when the number of generations G is set to 10 and the population scale is set to 50. We can conclude that higher fitness of individuals in the ISS scheme does not necessarily imply better steganographic security. Furthermore, from Fig. 7, we can conclude that PSNR is a better fitness function for the ISS compared to SSIM.

Table 1 The $E(oob)$ result of different population parameters with PSNR

n	G	Training set		Testing set		$E(oob)$
		PSNR	R	PSNR	R	
0(IMS)		55.35853	-	55.38898	-	0.4525
50	10	55.36424	4.1	55.39304	4.7	0.4648
	20	55.36463	1.7	55.39338	2.3	0.457
	30	55.36466	1.5	55.39375	1.3	0.4577
	40	55.36471	1.2	55.39384	1.1	0.4572
	50	55.36473	1	55.39388	1	0.4525
100	10	55.36431	4.5	55.39321	4.7	0.4571
	20	55.36464	3.6	55.39347	3.1	0.4573
	30	55.3647	2.8	55.39381	1.6	0.4528
	40	55.36479	1	55.39389	1.4	0.4531
	50	55.36479	1	55.39392	1	0.4547

Table 2 The $E(oob)$ result of different population parameters with SSIM

n	G	Training set		Testing set		$E(oob)$
		SSIM	R	SSIM	R	
0(IMS)		0.999869474	-	0.999873685	-	0.4525
50	10	0.999870188	7.5	0.999874042	8.3	0.4619
	20	0.999870251	2.6	0.999874246	4	0.4593
	30	0.999870305	1.7	0.999874307	2.8	0.4612
	40	0.999870316	1	0.999874389	1	0.4508
	50	0.999870316	1	0.999874389	1	0.4508
100	10	0.999870297	5	0.999874241	6.5	0.4589
	20	0.999870305	1.3	0.999874386	1.7	0.4558
	30	0.999870324	1	0.999874412	1	0.4534
	40	0.999870324	1	0.999874412	1	0.4534
	50	0.999870324	1	0.999874412	1	0.4534

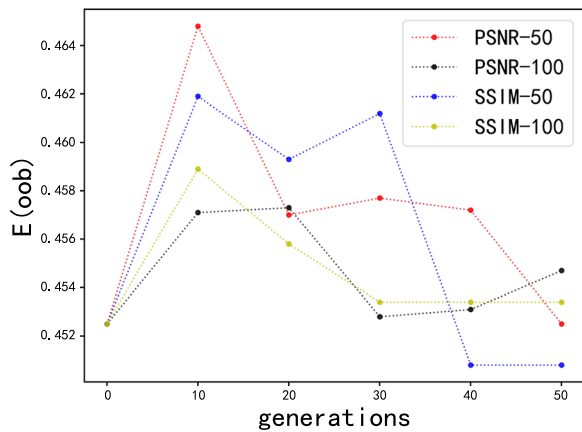


Fig. 7 The $E(oob)$ value in the condition of different generations numbers and fitness function

Table 3 The $E(oob)$ result of different image size and batch size

Image size	Batch size	Number of groups	$E(oob)$
512×512	256	18	0.4375
	144	32	0.4082
	64	72	0.3854
512×512	256	18	0.4375
256×256		72	0.418(0.4419)
128×128		288	0.3917(0.4706)

Optimizing image size and batch size

In this subsection, we randomly select 9216 images from the BossBase 1.01 database, with 4608 images for the training set and the remaining 4608 images for the testing set.

The experiments are divided into two parts.

In the first part, all images are kept to their original dimensions of 512×512 pixels, and batch sizes for the image sets of the ISS are set to 256, 144, and 64, respectively. As the total number of images in the experiment remains the same, the experimental groups corresponding to the three different batch sizes are 18, 32, and 72, respectively. The three experimental groups are tested by the SRM steganalysis, and the experimental results are given in Table 3.

In the second part, we investigate the effect of image size on steganographic security under the same batch size. In this experiment, the batch size is set to 256 images. Each image in the image set is further divided into three experimental control groups. The first group utilized the original image size of 512×512 pixels. The second and third groups divide all images into smaller blocks of sizes 256×256 pixels and 128×128 pixels,

respectively. In addition, to ensure that the number of pixels inputted into the SRM steganalysis remains the same, the experimental group with smaller image sizes should include a larger number of images. At the same time, we ensure that each group contained 256 images, regardless of their size. As a result, the experimental groups corresponding to the three different image sizes are 18, 72, and 288, respectively.

Then, we conduct the ISS multi-image steganography scheme with the same payload (0.2bpp) in each group of images. Additionally, we employ two different approaches to evaluate the impact of different image sizes on steganographic security using the SRM steganalysis. In the first approach, we directly utilize the small divided images directly as the training and testing set for steganalysis. In the second approach, we reassemble the divided small images and their corresponding stego images by concatenating them based on their positional information, and use these concatenated images as inputs for the SRM steganalysis method to evaluate the security of different steganography schemes.

The experimental procedure described above is depicted in Figs. 8 and 9, and the results are presented in Table 3. We can observe that when the image size remains constant, the security of the steganography scheme decreases as the batch size decreases. This is because smaller batch size limits the search space for the optimal image concatenation order in the ISS multi-image steganography scheme, thereby reducing the effectiveness of the ISS scheme.

Moreover, in the experiments focusing on image size, we discover that if we directly use the divided small images as training and testing data for the SRM steganalysis, the steganographic security improves as the image size decreases, as can be seen from the data in parentheses of Table 3. This is due to the fact that smaller images have fewer detectable features for SRM model, leading to a decrease in the performance of the steganalyzer. However, if we reassemble the divided small images and then apply the SRM steganalysis, the steganographic security deteriorates as the image size decreases, which aligns with our theoretical prediction.

In our experiments, we use the results from the second approach as the true indicator of steganographic security. This is because the second approach allowed us to place different experimental groups under exactly the same detection conditions, enabling a more objective and fair experimental results.

Therefore, we opt for a batch size of 256 images for the ISS scheme and conducted the multi-image steganography experiments using the original image size of 512×512 pixels.

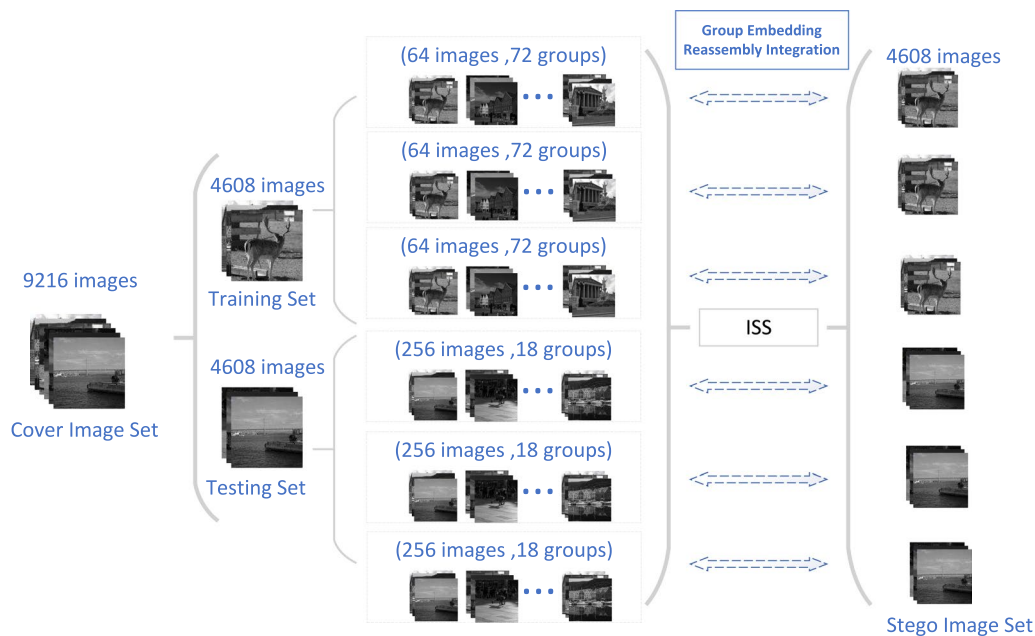


Fig. 8 ISS steganographic schemes based on different batch sizes

Security comparison among different multi-image steganography schemes

In this subsection, we randomly select 5120 images from the BossBase1.01 dataset and divide them into 10 groups, each consisting of 256 images for training and 256 images for testing. Next, we apply different multi-image steganography schemes to embed secret information in the images of each group.

In our experiment, we chose S-uniward, WOW, and HUGO as embedding schemes, and ISS, IMS, ES-DD, and UPD as multi-image steganography schemes. The payloads used in the multi-image steganography experiments are 0.1, 0.2, 0.3, 0.4, 0.5, and the steganographic security is evaluated based on $E(oob)$ as mentioned above. Figure 10 shows the comparative analysis of the security among different multi-image steganography schemes.

Based on the results presented in Table 4 and Fig. 10, it is evident that the ISS multi-image steganography scheme consistently exhibits the highest level of security compared to the other three schemes. Furthermore, when combined with non-additive embedding schemes, the ISS scheme shows a larger improvement in steganographic security. This is particularly evident in the results obtained with the HUGO-BD embedding scheme.

According to the result of our experiment, the ISS scheme enhances the security of steganography compared to the IMS scheme, which demonstrate the effectiveness of our efforts to optimize the image stitching sequence of cover images in multi-image steganography.

Additionally, the experimental results indicate that the IMS scheme and the ES-DD scheme exhibit similar performance in terms of security, which is consistent with the findings in (Liao et al. 2020). The UPD scheme has the same security as single-image steganography and exhibits the poorest performance in terms of security compared to the other three schemes. This finding suggests that the uneven distribution of payload among the images in multi-image steganography effectively contributes to enhancing the overall steganographic security. Among the three embedding schemes, the S-UNIWARD and WOW schemes exhibit similar levels of security, while the HUGO-BD scheme shows relatively lower security. This observation is consistent with their performance in single-image steganography.

The impact of different image merging schemes on the entropy of gray-level co-occurrence matrix

In order to explore why the ISS can achieve higher steganographic security compared to the IMS, we used the entropy of the Gray-Level Co-occurrence Matrix (GLCM) (Tamura et al. 1978) of images as an indicator to determine whether the merged image is a better carrier for steganography. As the entropy of the GLCM increases, it signifies a higher complexity in the texture of the image and a greater randomness in the distribution of pixels. Consequently, this indicates that the image is more suitable as a carrier for steganographic schemes. To optimize the utilization of GLCM entropy for representing the complexity of image textures and to minimize

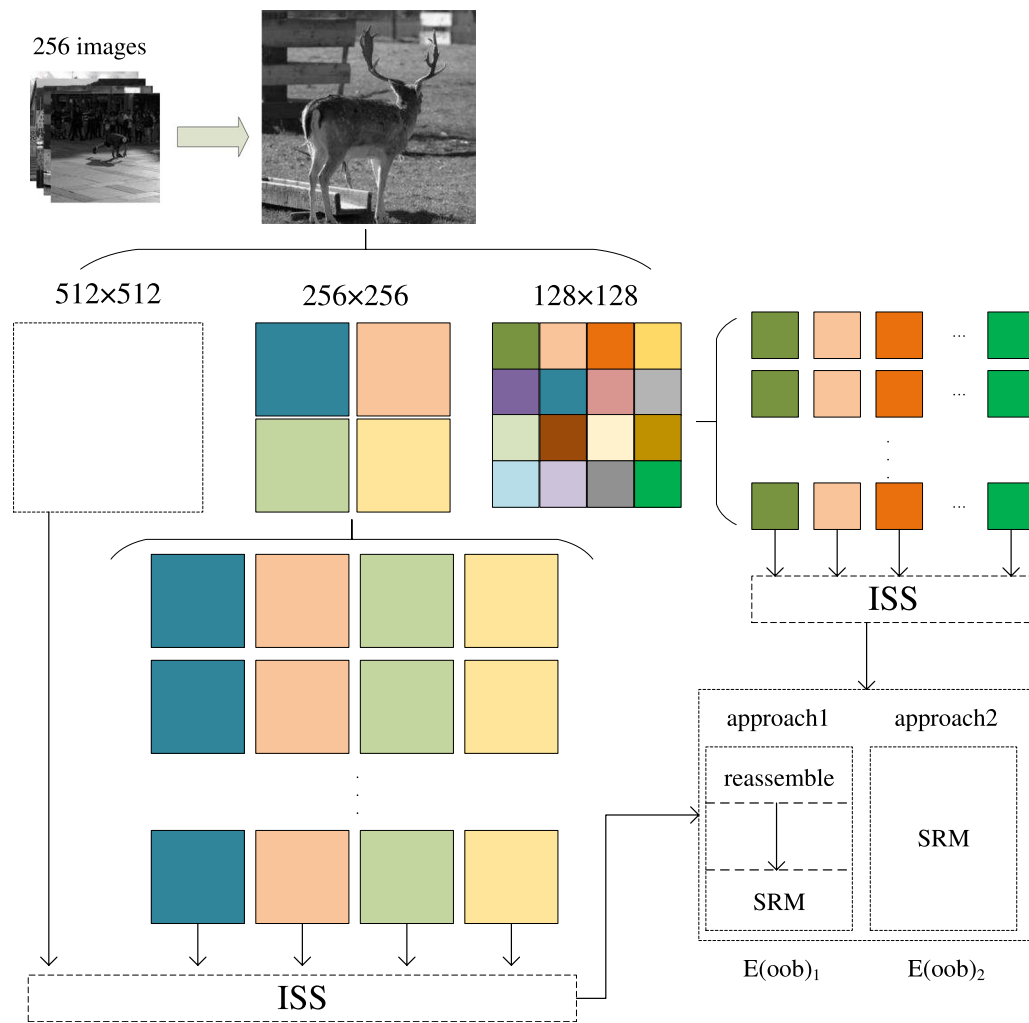


Fig. 9 ISS steganographic schemes based on different image sizes

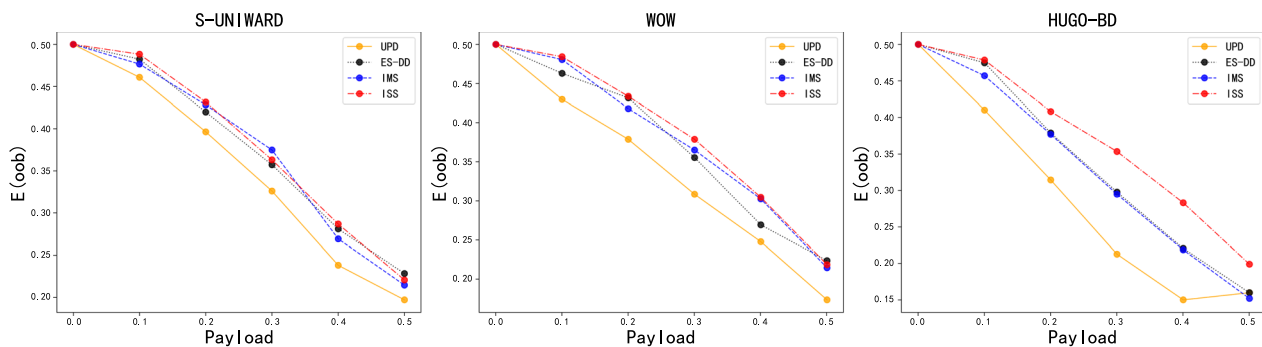


Fig. 10 Results for different payload distribution scheme

computational costs, we employ a grayscale level of 4 for the images and perform scanning in four directions: 0 degrees, 45 degrees, 90 degrees, and 135 degrees.

For the 10 sets of training images and 10 sets of testing images mentioned in Subsect. (4.3, we merge the images within each set using 10 randomly generated stitching sequence. Subsequently, we calculate the entropy of the

Table 4 Security comparison among different multi-image steganography schemes

Multi images steganography	Embedding algorithm	Payload				
		0.1	0.2	0.3	0.4	0.5
UPD	S-uniward	0.4609	0.3965	0.3262	0.2383	0.1973
ES-DD		0.4824	0.4199	0.3574	0.2813	<u>0.2285</u>
IMS		0.4766	0.4238	0.3633	0.2695	0.2148
ISS		<u>0.4883</u>	<u>0.4316</u>	<u>0.375</u>	<u>0.2871</u>	0.2209
UPD	HUGO-BD	0.3535	0.3145	0.2129	0.1504	0.1289
ES-DD		0.4746	0.3789	0.2978	0.2207	0.1602
IMS		0.457	0.377	0.2949	0.2188	0.1523
ISS		<u>0.4885</u>	<u>0.3828</u>	<u>0.3154</u>	<u>0.2354</u>	<u>0.1743</u>
UPD	WOW	0.4297	0.3789	0.3086	0.248	0.1738
ES-DD		0.4629	0.4316	0.3555	0.2695	<u>0.2239</u>
IMS		0.4805	0.418	0.3652	0.3027	0.2148
ISS		<u>0.4842</u>	<u>0.4336</u>	<u>0.3789</u>	<u>0.3047</u>	0.2188

Table 5 The impact of different image merging schemes on the entropy of GLCM

Image merging scheme	Embedding scheme	Entropy of GLCM
IMS		4.203597
ISS	S-uniward	4.203863
	WOW	4.204006
	HUGO-BD	4.203988

gray-level co-occurrence matrix for each merged image after each merging operation. Then we calculate the average of the 200 entropy values, which serves as the entropy of GLCM after merging the images using the IMS.

In the experiments conducted in Subsect. 4.3, for each set of images, the ISS is able to find an optimal image stitching sequence using a specific embedding scheme. Therefore, we can utilize the S-Uniward, WOW, and HUGO-BD schemes separately to obtain the entropy of the GLCM for each of the 20 image sets. Subsequently, we calculate the average entropy for each embedding scheme, which represents the entropy of the GLCM when using the ISS with a specific embedding scheme.

Table 5 presents a comparison of entropy of GLCM using IMS and ISS. It is evident that regardless of the embedding scheme employed, the ISS consistently yields higher entropy values compared to the IMS with a random stitching sequence.

We also conduct supplementary experiments to identify the stitching scheme that maximizes the GLCM entropy using optimization algorithms. The experimental setup follows Subsect. 4.3, using the S-UNIWARD embedding scheme with a payload of 0.2bpp. Before embedding the secret messages, we apply GA to optimize

Table 6 The experimental results of using GA to optimize the entropy of the GLCM

Schemes	Fitness	Training set	Testing set	E(oob)
ISS	PSNR	55.58726	55.12073	
	Entropy of GLCM	4.6407	4.7123	0.4316
GA-(Entropy of GLCM)	PSNR	55.58571	55.11804	
	Entropy of GLCM	4.6618	4.7311	0.4194

the entropy of the GLCM for the stitched images using the same dataset. The solution with the highest GLCM entropy is then used for embedding and steganalysis. The experimental results are shown in Table 6. However, our steganalysis tests revealed that the stitching method with the highest GLCM entropy did not necessarily exhibit the highest resistance to detection. Thus, we conclude that an increase in GLCM entropy is a necessary but not sufficient condition for enhancing the security of multi-image steganography.

Comparison of embedding change rates in different multi-image steganography schemes

In this subsection, we investigate the payload distribution among a batch of images for different multi-image steganography schemes under the same payload. To facilitate observation, we select 100 images from a batch of images and calculate the pixel change rates for their corresponding cover images and steganographic images.

From Fig. 11, it is evident that the UPD scheme exhibits a uniform payload distribution across all images during multi-image steganography. This indicates that the UPD scheme treats all images equally when embedding secret

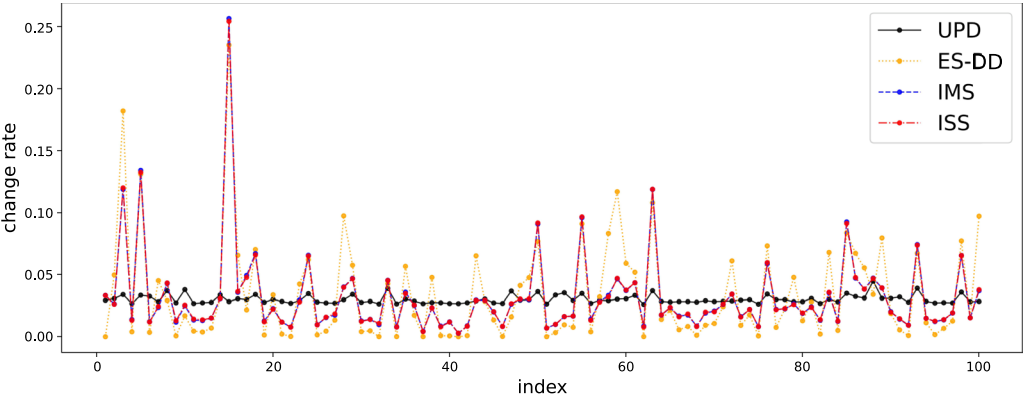


Fig. 11 Comparison of embedding change rates in different multi-image steganography schemes

messages. Unlike the UPD scheme, the other three multi-image steganography schemes exhibit non-uniform payload distribution.

In the case of the ES-DD scheme, it is noteworthy that a significant number of images have a change rate of 0, implying that during the embedding of secret information, some images are intentionally skipped while retaining their original cover image. Furthermore, when compared to the other three schemes, the ES-DD scheme exhibits a more significant variation in pixel change rates among different images. Moreover, the pixel change rates caused by the embedding process of the ISS scheme and the IMS scheme are quite similar.

Comparison of different optimization algorithms

To evaluate the performance of other optimization algorithms in conjunction with the multi-image steganography, we assess three popular optimization algorithms: the Rime Optimization Algorithm (RIME) (Su et al. 2023), the Sparrow Search Algorithm (SSA) (Xue and Shen 2020) and the Whale Optimization Algorithm (WOA) (Mirjalili and Lewis 2016). For each algorithm, the initial population size is set to 50, and 50 iterations are conducted. Each experiment optimize the image stitching scheme by randomly selecting 256 grayscale images from

the BossBase v1.01 database. The embedding scheme used is S-UNIWARD, with the payload set to 0.2 bpp. The PSNR of the cover image and stego image continues to be chosen as the fitness function for the four optimization algorithms. The average results from 10 trials are summarized in Table 7.

It can be observed that among the four optimization algorithms, the SSA achieves convergence with the fewest iterations and has the shortest runtime. Once these algorithms converge, the WOA exhibits the highest fitness. However, after 50 iterations, the GA shows the greatest resistance to detection in SRM steganalysis. We can conclude that optimization algorithms that converge faster and achieve higher fitness do not necessarily ensure better resistance to detection. Similar results can also be observed in Subsect. (4.1, the GA individuals we ultimately selected are not necessarily those with the highest fitness. This is because PSNR, as a fitness function, does not fully reflect steganographic security; it only demonstrates the highest resistance to detection when PSNR reaches a certain threshold. Therefore, in our work, the metrics traditionally used to evaluate the performance of optimization algorithms (such as convergence speed and fitness level) are not applicable. As long as the optimization algorithm iterates in the correct

Table 7 Performance of the four optimization algorithms

Optimization algorithm	RIME	SSA	WOA	GA
Common parameter	Initial population:50 iterations:50			
customizable parameters	-	-	-	Selection, crossover and mutation operators; Crossover and mutation probabilities
Number of iterations at convergence	21.2	15.7	43.6	35.4
Runtime	7h53m	5h26m	9h25m	8h36m
Fitness after convergence	54.39415	54.39509	54.39718	54.39284
E(oob)	0.4459	0.433	0.4483	0.4507

direction, we can identify the individual with the strongest resistance to detection during the iteration process by fine-tuning parameters such as the algorithm's iteration speed, its capacity to escape local optima, and other relevant parameters. As can be seen from Table 7, the GA provides more customizable parameters compared to other optimization algorithms, making it better suited to meet our needs.

Conclusions

In this paper, we propose a novel multi-image steganography scheme ISS. It utilizes genetic scheme to solve the problem of local optimal stitching scheme for the cover images, thereby achieving an optimal payload distribution scheme for multi-image steganography. The experimental results indicate that, in most cases, the ISS scheme exhibits stronger resistance to detection compared to the current state-of-the-art multi-image steganography schemes IMS and ES-DD.

Furthermore, we observe that our scheme fails to find a global optimal solution for the image stitching scheme that could effectively enhance steganographic security. Instead, it converges to a local optimal solution, primarily due to two reasons. Firstly, the GA has proven to be inefficient as an optimization method. In future research, we will explore more cost-effective optimization techniques. Secondly, the ISS scheme uses PSNR as the fitness function, but PSNR cannot directly indicate the level of steganographic security. In the future, we plan to explore better fitness functions that are more closely related to steganographic security, to improve the scheme's efficiency and generate more secure stego images.

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Author contributions

Shihao Zhang: Conceptualization; Original Draft Preparation; Writing & Editing; Yanhui Xiao: Investigation; Writing; Supervision; Huawei Tian: Conceptualization; Original Draft Preparation; Xiaolong Li: Writing; Supervision

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Availability of data and materials

Some data used in this paper were obtained from the publicly accessible websites (https://dde.binghamton.edu/download/stego_algorithms/). Other data are available from corresponding authors.

Declarations

Competing interests

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work

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References

- Bernard S, Bas P, Klein J, Pevn T (2020) Explicit optimization of min max steganographic game. *IEEE Trans Inf Forens Secur* 16:812–823
- Bernard S, Bas P, Pevn T, Klein J (2021) Optimizing additive approximations of non-additive distortion functions. In: *Proceedings of the 2021 ACM workshop on information hiding and multimedia security*, pp 105–112
- Bernard S, Pevn T, Bas P, Klein J (2019) Exploiting adversarial embeddings for better steganography. In: *Proceedings of the ACM workshop on information hiding and multimedia security*, pp 216–221
- Cheddad A, Condell J, Curran K, Mc Kevitt P (2010) Digital image steganography: survey and analysis of current methods. *Signal Process* 90(3):727–752
- Cogranne R, Sedighi V, Fridrich J (2017) Practical strategies for content-adaptive batch steganography and pooled steganalysis. In: *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pp 2122–2126
- Denemark T, Fridrich J (2015) Improving steganographic security by synchronizing the selection channel. In: *ACM*, pp 5–14
- Duan X, Li B, Yin Z, Zhang X, Luo B (2023) Robust image steganography against lossy jpeg compression based on embedding domain selection and adaptive error correction. *Expert Syst Appl* 229:120416
- Fridrich J, Filler T (2007) Practical methods for minimizing embedding impact in steganography. In: *Security, steganography, and watermarking of multimedia contents IX*, vol 6505. *SPIE*, pp 13–27. <https://doi.org/10.1117/12.697471>
- Filler T, Fridrich J (2010) Gibbs construction in steganography. *IEEE Trans Inf Forens Secur* 5(4):705–720
- Fridrich J, Goljan M, Soukal D (2004) Perturbed quantization steganography with wet paper codes. In: *Proceedings of the 2004 workshop on multimedia and security*, pp 4–15
- Filler T, Judas J, Fridrich J (2011) Minimizing additive distortion in steganography using syndrome-trellis codes. *IEEE Trans Inf Forens Secur* 6(3):920–935. <https://doi.org/10.1109/TIFS.2011.2134094>
- Fridrich J, Kodovsky J (2012) Rich models for steganalysis of digital images. *IEEE Trans Inf Forens Secur* 7(3):868–882. <https://doi.org/10.1109/TIFS.2012.2190402>
- Fridrich J, Kodovsky J (2013) Multivariate gaussian model for designing additive distortion for steganography. In: *2013 IEEE international conference on acoustics, speech and signal processing*. *IEEE*, pp 2949–2953
- Holub V, Fridrich J (2012) Designing steganographic distortion using directional filters. In: *2012 IEEE international workshop on information forensics and security (WIFS)*. *IEEE*, pp 234–239
- Holub V, Fridrich J, Denemark T (2014) Universal distortion function for steganography in an arbitrary domain. *EURASIP J Inf Secur* 2014(1):1
- Ker AD (2006) *Batch steganography and pooled steganalysis*. Springer, Berlin
- Ker AD (2007) Batch steganography and the threshold game. In: *Security, steganography, and watermarking of multimedia contents IX*, vol 6505. *SPIE*, pp 41–53
- Ker AD (2007) A capacity result for batch steganography. *IEEE Signal Process Lett* 14(8):525–528. <https://doi.org/10.1109/LSP.2006.891319>
- Ker AD (2008) Perturbation hiding and the batch steganography problem. In: *Information hiding: 10th international workshop, IH 2008, Santa Barbara, CA, USA, May 19–21, 2008, Revised Selected Papers 10*. Springer, pp 45–59
- Khan IH (2015) Assessing different crossover operators for travelling salesman problem. *Int J Intell Syst Appl* 7(11):19–25
- Kanan HR, Nazeri B (2014) A novel image steganography scheme with high embedding capacity and tunable visual image quality based on a genetic algorithm. *Expert Syst Appl* 41(14):6123–6130
- Ker AD, Pevn T (2012) Batch steganography in the real world. In: *Proceedings of the on multimedia and security*. pp 1–10
- Ker AD, Pevn T (2014) The steganographer is the outlier: realistic large-scale steganalysis. *IEEE Trans Inf Forens Secur* 9(9):1424–1435

- Li B, Wang M, Huang J, Li X (2014) A new cost function for spatial image steganography. In: 2014 IEEE international conference on image processing (ICIP). IEEE, pp 4206–4210
- Li B, Wang M, Li X, Tan S, Huang J (2015) A strategy of clustering modification directions in spatial image steganography. *IEEE Trans Inf Forensics Secur* 10(9):1905–1917
- Liao X, Yin J, Chen M, Qin Z (2020) Adaptive payload distribution in multiple images steganography based on image texture features. *IEEE Trans Dependable Secure Comput* 19:897–911
- Martín A, Hernández A, Alazab M, Jung J, Camacho D (2023) Evolving generative adversarial networks to improve image steganography. *Expert Syst Appl* 222:119841
- Mirjalili S, Lewis A (2016) The whale optimization algorithm. *Adv Eng Softw* 95:51–67
- Pevný T, Filler T, Bas P (2010) Using high-dimensional image models to perform highly undetectable steganography. *Lect Notes Comput Sci* 6387:161–177
- Qin X, Li B, Tan S, Zeng J (2019) A novel steganography for spatial color images based on pixel vector cost. *IEEE Access* 7:8834–8846
- Rustad S, Andono PN, Shidik GF et al (2022) Digital image steganography survey and investigation (goal, assessment, method, development, and dataset). *Signal Process* 206:108908
- Sedighi V, Cogranne R, Fridrich J (2015) Content-adaptive steganography by minimizing statistical detectability. *IEEE Trans Inf Forensics Secur* 11(2):221–234
- Steganalysis HCDB, Westfeld A (2001) F5-a steganographic algorithm. In: *Information hiding: 4th international workshop, IH 2001, Pittsburgh, PA, USA, April 25–27, 2001. Proceedings*, vol 2137. Springer, p 289
- Su H, Zhao D, Heidari AA, Liu L, Zhang X, Mafarja M, Chen H (2023) RIME: a physics-based optimization. *Neurocomputing* 532:183–214
- Tan J, Liao X, Liu J, Cao Y, Jiang H (2021) Channel attention image steganography with generative adversarial networks. *IEEE Trans Netw Sci Eng* 9(2):888–903
- Tang W, Li B, Luo W, Huang J (2015) Clustering steganographic modification directions for color components. *IEEE Signal Process Lett* 23:197–201
- Tang W, Li B, Tan S, Barni M, Huang J (2018) CNN based adversarial embedding with minimum alteration for image steganography. *arXiv preprint arXiv:1803.09043*
- Tamura H, Mori S, Yamawaki T (1978) Textural features corresponding to visual perception. *IEEE Trans Syst Man Cybern* 8(6):460–473. <https://doi.org/10.1109/TSMC.1978.4309999>
- Volkhonskiy D, Nazarov I, Burnaev E (2020) Steganographic generative adversarial networks. In: *Twelfth International Conference on Machine Vision (ICMV 2019)*, vol 11433. SPIE, pp 991–1005
- Wang Y, Zhang W, Li W, Yu X, Yu N (2020) Non-additive cost functions for color image steganography based on inter-channel correlations and differences. *IEEE Trans Inf Forensics Secur* 15:2081–2095
- Xue J, Shen B (2020) A novel swarm intelligence optimization approach: sparrow search algorithm. *Systems Sci Control Eng* 8(1):22–34
- Zhao Z, Guan Q, Zhao X, Yu H, Liu C (2018) Universal embedding strategy for batch adaptive steganography in both spatial and jpeg domain. *Multimed Tools Appl* 77:14093–14113
- Zhang Y, Zhang W, Chen K, Liu J, Liu Y, Yu N (2018) Adversarial examples against deep neural network based steganalysis. In: *Proceedings of the 6th ACM workshop on information hiding and multimedia security*, pp 67–72
- Zhang W, Zhang X, Wang S (2008) Maximizing steganographic embedding efficiency by combining hamming codes and wet paper codes. In: *Information hiding: 10th international workshop, IH 2008, Santa Barbara, CA, USA, May 19–21, 2008, Revised Selected Papers* 10. Springer, pp 60–71

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